

Course: Data Science

Course Code: 3031053

Prerequisite: Basic Knowledge of Programming

Rationale : This course provides necessary knowledge on data manipulation and to perform analysis on the practical problems using statistical and machine learning approach to generate report and visualize the results in graphical form using programming tool.

Teaching and Examination Scheme										
Teaching Scheme					Examination Scheme					
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	0	0	3	20	20	-	60	-	100

SEE - Semester End Examination, T - Theory, P - Practical

Course Content				
Sr.	Topics	W	T	
1	INTRODUCTION: Data Science: Introduction to Data Science – Digital Universe – Sources of Data – Information Commons – Data Science Project Life Cycle: OSEMN Framework.	9	4	
2	DATA PREPROCESSING: Introduction to Data Preprocessing – Reading, Selecting, Filtering Data – Filtering Missing Values – Manipulating, Sorting, Grouping, Rearranging, Ranking Data.	14	6	
3	CONCEPT LEARNING: Formulation of Hypothesis – Probabilistic Approximately Correct Learning – VC Dimension – Hypothesis elimination – Candidate Elimination Algorithm.	16	7	
4	BASICS OF R: R Basics – data types and objects – control structures – data frame – Feature Engineering – scaling, Label Encoding and One Hot Encoding, Reduction.	19	8	
5	MODEL FITTING IN R: Regression Models – Linear and Logistic Model, Classification Models – Decision Tree, Naïve Bayes, SVM and Random Forest, Clustering Models – K Means and Hierarchical clustering.	19	9	
6	VISUALIZATION: Data visualization: Box plot, histogram, scatter plot, heat map – Working with Tableau – Outlier detection – Data Balancing.	14	7	
7	PERFORMANCE EVALUATION in R: Loss Function and Error: Mean Squared Error, Root Mean Squared Error – Model Selection and Evaluation criteria: Accuracy, Precision, F1 score, Recall Score – Binary Predictive Classification – Sensitivity, Specificity.	9	5	
		100	46	

W - Weightage (%) , T - Teaching hours

Reference Books

1.	Introduction to Machine Learning, Fourth Edition, MIT Press, 2020 (TextBook) By Ethem Alpaydin
2.	R for Data Science: Import, Tidy, Transform, Visualize, and Model Data, 2017 By Hadley Wickham, Garrett Golemund
3.	Data Mining: Concepts and Techniques, Morgan Kaufmann, 2011 By Han, J., Kamber, M., Pei, J.
4.	An Introduction to Statistical Learning with Applications in R, Springer, 2013 By James, G., Witten, D., Tibshirani, R.

Course Outcome

After Learning the Course the students shall be able to:	
1.	Ability to gain basic knowledge on data science.
2.	Convert the real time data into suitable form for analysis.
3.	Gain the insights from the data through statistical inferences.
4.	Develop suitable models using machine learning techniques and to analyze its performance.
5.	Identify the requirement and visualize the results.
6.	Analyze on the performance of the model and the quality of the results.